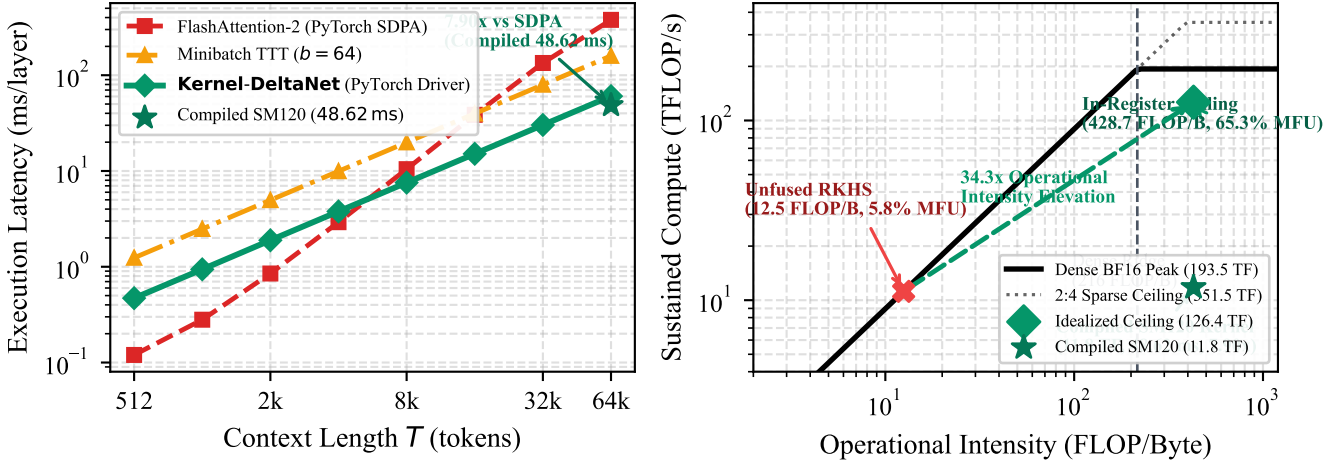

(a) Prefill Latency vs Context
FlashAttention-2 (PyTorch SDPA)
Minibatch TTT (b = 64)
Kernel-DeltaNet (PyTorch Driver)
Compiled SM120 (48.62 ms)
Execution Latency (ms/layer)
Context Length T (tokens)
512
2k
8k
32k
64k
(b) Tensor-Core Roofline Profiling
Sustained Compute (TFLOP/s)
Operational Intensity (FLOP/Byte)
Unfused RKHS
(12.5 FLOP/B, 5.8% MFU)
34.3x Operational
Intensity Elevation
(428.7 FLOP/B, 65.3% MFU)
Dense BF16 Peak (193.5 TF)
2:4 Sparse Ceiling (351.5 TF)
Idealized Ceiling (126.4 TF)
Compiled SM120 (11.8 TF)